YAMA BOT – Requirements (v3)

Hotel Competitor Room Mapping & Price Monitoring

1. Project purpose

YAMA Bot uses a Booking.com API provider on RapidAPI to collect competitor hotel rooms and their prices. The system has two separate workflows: a one-time setup to establish hotel and room mappings, followed by repeatable price extraction using those saved mappings. One input workbook and one dated output report are used per hotel.

2. Hotel input workbook

Example: MSP D'Spark Hotel Bayu.xlsx. Each hotel has its own workbook with the following sheets:

Sheet	Purpose
hotel_data	Our hotel name, room types, room names, minimum and maximum rates. Prepared manually.
competitor_hotel	Competitor hotel IDs and details retrieved during initial setup.
competitor_room	Competitor room IDs and names, gross_amount, and manually assigned mapping to our room_type.
competitor_price	Formatted report template to be copied into each dated output workbook.

3. Flow 1 – Initial hotel and room mapping setup

Run once for each hotel during onboarding. Run again only when hotel/competitor details or room mappings need changes.

Step 1 – Enter the hotel's room types and rate limits manually in hotel_data.

Step 2 – Find each competitor using RapidAPI GET Search Hotel Destination (parameters: hotel name in booking.com). Save the competitor hotel ID.

Step 3 – Use GET Get Hotel Details (parameters: hotel_id, arrival_date, departure_date) to fill competitor_hotel information.

Step 4 – Use GET Get Room List (parameters: hotel_id, arrival_date, departure_date) to obtain competitor room IDs, room names and gross_amount.

Step 5 – Review the competitor rooms and manually map each suitable competitor room to our room_type in competitor_room. Save the input workbook.

4. Flow 2 – Recurring competitor price extraction

Run whenever updated competitor prices are required. It uses the existing mapping from Flow 1; it does not repeat hotel discovery or automatically change room mappings.

Step 1 – Open the hotel input workbook and read saved competitor_hotel IDs and mapped competitor_room IDs / room_type values.

Step 2 – Before making price API calls, create a dated workbook in the output folder. Copy only the competitor_price sheet and competitor_room sheet from input workbook into that output workbook. Keep the input workbook unchanged. Or use Google Sheets and save in Google Drives. After then forward the workbook to Telegram Account that will be created later.

Step 3 – Call RapidAPI GET Get Room List (parameters: hotel_id, arrival_date, departure_date) for each competitor hotel represented by the saved mappings.

Step 4 – Use only API records whose competitor_room_id matches an existing mapped room in competitor_room. Read the latest gross_amount for those rooms.

Step 5 – Write retrieved gross_amount values into the corresponding mapped rows on the output competitor_room sheet, and write the matching competitor prices into the output competitor_price sheet.

Step 6 – For each of our room types, calculate the average and lowest valid competitor gross_amount and write them into the competitor_price sheet report workbook. Save the dated output workbook.

5. Output workbook

Example: 2026-10-26-MSP D'Spark Hotel Bayu.xlsx

Output sheet	Contents
competitor_room	Copied mapped room table, updated with latest gross_amount for successfully matched rooms.
competitor_price	Copied hotel report template, populated with corresponding mapped competitor prices and average/lowest prices for each room type.

6. Simple processing rules

- Only competitor rooms already mapped in the input workbook are included in the price report.
- Match API room records using competitor hotel ID and competitor room ID, not name similarity.
- Missing, unavailable, or invalid prices are left blank/marked N/A and excluded from average and lowest calculations; never treat missing prices as zero.
- The original input workbook is not overwritten during recurring extraction.
- The number of competitors and hotel room types may vary from hotel to hotel.
- Arrival and departure dates must be specified for each API price-extraction run.

7. Deliverable

For each hotel: a completed input workbook containing saved manual mappings, and a dated output workbook containing only competitor_room and competitor_price sheets with the latest mapped competitor rates and their calculated average and lowest values.